Institute for Marine and Antarctic Studies

HDR Project Approval Form

1. Project details			
Proposed degree	PhD	~~MRes~~	
Proposed program	~~IMAS~~	QMS	AAPP
Project title	Prognostic modeling of Antarctic landfast sea ice in a coupled regional ocean-sea ice-atmosphere model		
Closing Date	6 months	~~1 year~~	

Project outline & methods

Please provide an overview of the project, a brief research plan and an explanation of the methods to be applied. ***For QMS projects****, emphasize the quantitative nature of the project over and above what is expected of a standard IMAS project.*

Background

Landfast sea ice (fast ice) is a key component of the Antarctic cryosphere. Its distribution strongly influences the size, shape and sea ice production of coastal polynyas (Fraser et al., 2019), thus modifying the physical ocean properties on the continental shelf (e.g., Ohshima et al., 2013) as well as modifying the momentum transfer from the atmosphere to the ocean (Kusahara et al., 2020), yet its current representation in coupled regional models is rudimentary (i.e., prescribed, or ignored completely). Indeed, it has been identified as a "missing process" in the current generation of regional models. Coupled sea ice-ocean models which ignore fast ice produce dense water in the wrong regions (Fraser et al., 2019), and probably in the wrong quantities (Kusahara et al., in prep.). Thus, its development into coupled models is a strong focus, and will enable much more accurate modeling of coastal water mass modification, without resorting to prescribed surface heat/salt flux forcing.

Recent work at UTAS/ACE CRC has produced the first high-quality, long time-series dataset of Antarctic fast ice extent (Fraser et al., 2020). While this presents a significant advance in terms of characterisation of the Antarctic nearshore coastal advance, it is only a stepping-stone toward full understanding of the wider effects of fast ice. Complete incorporation of fast ice into a model environment is required to achieve this understanding.

This limitation has long been recognised in the Arctic (e.g., Adams et al., 2011), and steps have been taken to address it. Researchers working in the Arctic have recently worked toward including fast ice in their models by a) grounding ice upon its own keels when sufficiently deep; and b) modifying ice rheology to include a tensile strength component (Lemieux et al., 2016). This combination of parameterisations has been shown to accurately represent the long-term mean distribution of fast ice in the Arctic (Lemieux et al., 2016), especially when a third element of tidal forcing is included (Lemieux et al., 2018). The Lemieux parameterisation of tensile strength is now a part of CICE version 6.

By contrast to Arctic fast ice which grounds upon its own keels in the relatively shallow bathymetry, Antarctic fast ice extent is closely related to the distribution of grounded icebergs, from which mechanical stability is imparted to the fast ice. This is due to a combination of much deeper Antarctic continental shelf bathymetry plus the presence of icebergs which can ground upon shoals of up to ~400 m depth.

Thus, with an accurate dataset of grounded icebergs plus the addition of tensile strength to the sea ice rheology, it is possible to model Antarctic fast ice. A PhD project at

UCLouvain, Belgium, is currently using the Lemieux parameterisation of fast ice within the NEMO-LIM coupled ocean-sea ice framework to investigate ocean/ice interaction north of the Totten and Moscow University Ice Shelves, East Antarctica. This is the first model incorporating prognostic Antarctic fast ice. This project is only in early stages, but has shown a high degree of sensitivity of fast ice distribution to the (prescribed) grounded iceberg distribution. The UCLouvain study is limited in geographical scope, leaving an opening for other groups to fill this space (i.e., to model other regions).

The candidate will be heavily involved in geophysical model development, tuning and running, including incorporation of the Lemieux tensile strength parameterization into the sea ice model. As such, this project is particularly quantitative in nature, over-and-above the level expected in a standard IMAS PhD project.

Aims/Objectives

The Australian Bureau of Meteorology (BoM) has recognised the importance of incorporating realistic fast ice into its regional coupled modeling work, and aims to develop the capability to prognostically model fast ice to achieve this. BoM will work within a coupled regional NEMO-SI3 framework to achieve this. **PHIL - what resolution? Which atmosphere? What forcing methodology – e.g., repeated annual? Etc. – Please add as much detail as you like**

Methods

The successful PhD student will:

- Modify SI3 to include the Lemieux tensile strength parameterisation plus basal stress from grounded icebergs (N.B. both Jean-Francois Lemieux (Environment Canada) and Thierry Fichefet (UCLouvain) are collaborators and can assist with this).
- Develop a dataset of grounded iceberg distribution based on repeat SAR imagery in the region/s of interest.
- Run and tune the model to produce realistic fast ice (assessing distribution against the Fraser et al., 2020 dataset), in a region including both “regime 1” (dynamically-formed) and “regime 2” (thermodynamically-formed) fast ice (as defined in Fraser et al., 2012).
- Perform a series of sensitivity tests to determine the primary environmental drivers of fast ice extent.
- Perform a series of experiments to quantify the effects of realistic fast ice on the ocean, atmosphere and sea ice – including the formation of sea ice in coastal polynyas, and the subsequent production of dense shelf water and Antarctic bottom water. This may include a comparison against a model run with fixed, prescribed fast ice (as in Kusahara et al., 2017).
- Assess modeled water mass transformation performance against observations (ship- and mammal-based Conductivity-Temperature-Depth profiles; mooring data).

Proposed Chapters

1) **Modeling realistic Antarctic landfast ice by adding tensile strength and grounded icebergs**
2) **The sensitivity of Antarctic landfast ice extent to environmental forcing**

3) Effects of realistic and dynamic fast ice on the formation of Antarctic Bottom Water in the Cape Darnley Polynya, East Antarctica

References:

Fraser, A. D., R. A. Massom, K. J. Michael, B. K. Galton-Fenzi, and J. L. Lieser, 2012: East Antarctic Landfast Sea Ice Distribution and Variability, 2000–08. J. Climate, 25, 1137–1156, https://doi.org/10.1175/JCLI-D-10-05032.1.

Kusahara, K., H. Hasumi, A. D. Fraser, S. Aoki, K. Shimada, G. D. Williams, R. Massom, and T. Tamura, 2017: Modeling Ocean–Cryosphere Interactions off Adélie and George V Land, East Antarctica. J. Climate, 30, 163–188, https://doi.org/10.1175/JCLI-D-15-0808.1.

<Others – this is still incomplete sorry!>

Summary for advertising

Please provide a short summary of the proposed project, suitable for advertising on the web. You should also include the required and desirable skills, qualifications and experience needed for the project. Projects need to be advertised for a minimum of 4 weeks before a candidate can be allocated unless a fast-track exemption is approved.

Summary	
Essential skills/experience	1. Experience with fluid dynamics, especially in a geophysical modeling context (ocean, atmosphere or other fluid modeling) 2. Strong analytical skills, especially with scientific analysis/programming software (e.g., Python, Matlab)
Desirable skills/experience	1. Experience using supercomputer and related infrastructure (e.g., linux skills) 2. Ability to develop research-independence within this project (i.e., to "take ownership" of the project)
Assessment criteria	Applicants will be assessed and ranked according to the quality of their basis for entry research degree and institution, prior peer reviewed publications, academic awards, project-specific skills, training or relevant industry experience, referee's reports and supervisory support.
Fast track advertisement? Two week period	~~Yes~~ No

Justification

Please provide a justification for advertisement exemption based on the suitability and exceptional academic credentials of an available candidate, timing, intellectual property protection etc

2. Supervisory Team - *See Appendix 1 for criteria*

Primary Supervisor *(must be IMAS contract/continuing staff member)*				
Name	Alexander Fraser			
Centre	O&C			
Area of expertise	Antarctic landfast sea ice; sea ice remote sensing			
Supervisory fraction	25 %			
Current load	3	Primary supervised students	1	Co-supervised students
Experience	10	Years supervisory experience	2	Completed students
Co-Supervisor 1				
Name	Will Hobbs			
Centre / Organisation	O&C/AAPP			
Contact if external	Email:		Phone:	
Area of expertise	Antarctic sea ice variability; regional and global model analysis			
Supervisory fraction	20 %			
Current load	WILL	Primary supervised students	WILL	Co-supervised students
Experience	WILL	Years supervisory experience	WILL	Completed students
Co-Supervisor 2				
Name	Phil Reid			
Centre / Organisation	BoM			
Contact if external	Email: Phillip.Reid@bom.gov.au		Phone:	
Area of expertise	Sea ice/ocean/atmosphere interaction modelling and analysis			
Supervisory fraction	20 %			
Current load	PHIL	Primary supervised students	PHIL	Co-supervised students
Experience	PHIL	Years supervisory experience	PHIL	Completed students
Co-Supervisor 3				
Name	Paul Sandery			
Centre / Organisation	CSIRO Hobart			
Contact if external	Email: Paul.Sandery@csiro.au		Phone:	
Area of expertise	Climate modeling			
Supervisory fraction	15 %			
Current load	PAUL	Primary supervised students	PAUL	Co-supervised students
Experience	PAUL	Years supervisory experience	PAUL	Completed students
Research Advisor				
Name	Wilma Huneke			
Position	Ocean modeler			
Centre / Organisation	ANU			
Contact if external	Email:	Phone:		
Area of expertise	Antarctic regional coastal sea ice/ocean interaction modelling			
Supervisory fraction	10 %			
Research Advisor				
Name	Gary Brassington OR Stewart Allen			
Position	Ocean modeler			
Centre / Organisation	BoM			
Contact if external	Email:	Phone:		
Area of expertise	Ocean modeling			
Supervisory fraction	10 %			

Justification

Provide a justification for supervisory team members based on the suitability of the supervisory teams' academic expertise, supervisory experience, likelihood of supervision throughout candidature and supervisory performance record.

Please explain how the supervisory team meets the strategic goals of equality, diversity of career stage, development of ECRs or newly hired academics and appropriateness of supervisory loads.
Academic expertise: Fraser is a mid-career researcher and a world-leader in studies on Antarctic fast, including mapping and studies of its variability. OTHERS **Supervisory experience:** Fraser: 2 PhD completed, 4 current. OTHERS **Likelihood of supervision throughout candidature:** Fraser: Medium (contract expires April 2023 – possible 1 yr ext). OTHERS **Supervisory performance record:** Fraser: 2 PhD completions; both within 4 year max. Experience with challenging candidates. OTHERS **Equality, diversity of career stage, development of ECRs:** TBC

3. Budget & Infrastructure

Project costs *Provide details of costs required to complete the project including: equipment, laboratory costs, data costs, field work costs, computing resources over and above basic desktop/laptop computer, any training required, etc. Project costs could be from competitive grants, for example, ARC Discovery/Linkage grants, AAPP, Blue Economy CRC, or other sources, such as industry, Government agencies.*	
Project cost details	
Total project cost $	
Co-funding *Please state the resources available to this PhD project including the project account name and number against which the project costs should be coded.*	
Available research project $ *Project name and code*	
Student operating funds $	$5,250 ($1,500 per year for 3.5 years)
Variance	
Scholarship *Please indicate if resources are available to **fund or co-fund a scholarship**, for example from a grant, from industry, or from the student's home country.*	
Scholarship source	AAPP
Scholarship $	Equivalent to RTP
Top-up scholarship (pre-award)	AAPP, $6k per annum
Infrastructure Usage Please provide any extra information that will help in establishing the infrastructure needs of the project, such as the type of laboratory required, other field and laboratory equipment, sharing arrangements, or timing of usage.	
Lab space	N/A
Aquarium space	N/A
Histology & microscopy	N/A
Vehicles & Vessels	N/A

Diving Equipment	N/A
Desk location	Waterfront
Storage (incl. quarantine – fridges/freezers)	N/A

External infrastructure
If any of the infrastructure needs are required from organisations (e.g. CSIRO, AAD) or schools (e.g. School of Natural Sciences) external to IMAS please provide a memo of agreement from the relevant organisation/department.

BoM – supercomputer
(Or NCI – I'll leave it to you Phil!)

4. Strategic alignment

IMAS Research Priority Area(s)
Please nominate one or more IMAS Research Priority Areas within which this project fits. Please refer to The Institute for Marine and Antarctic Studies Strategic Plan 2017–2022, http://www.imas.utas.edu.au/__data/assets/pdf_file/0020/1021385/IMAS-Strategic-Plan-FINAL-low-res.pdf

- ☐ Fisheries Assessment
- ☐ Food Security Innovation
- ☐ Whole of Ecosystem Management
- ☐ Ocean and Antarctic Governance
- ☒ Ocean and Antarctic physics and Chemistry
- ☐ Ecology and Ecosystem Dynamics
- ☐ Biodiversity and Marine Habitats
- ☐ Marine Socio-ecology
- ☒ Cryosphere and Solid Earth
- ☒ AAPP

AAPP: nominate one or more of the 7 AAPP projects within which the HDR project fits and provide a brief statement how it aligns with the AAPP objective.

Addresses a defined deliverable for Project 6: Sea ice:

"Continue multi-variable analysis using new observations and updated numerical simulation including focus on Antarctic fast-ice, polynya and BGC processes. [P1, P3, P4, P5, P7]"

Please provide a brief statement on how this project aligns with the IMAS Research Priority Areas:

TBC

5. Approvals

Project Management	
Primary Supervisor Signature	
QMS Committee signature send to: Christine.Fury@utas.edu.au	
AAPP Program Leader signature: send to Wenneke.tenhout@utas.edu.au	
Graduate Research Coordinator (GRC to return to primary supervisor if needed)	
iGrad checked for overdue actions?	
Does the supervisory team have suitable academic expertise, supervisory experience, supervisory performance record and probability of providing appropriate supervision for the lifetime of the PhD project?	
Does the team structure meet IMAS goals in terms of diversity (e.g. considering development of ECRs and new academics)?	
GRC Signature	
Lead GRC signature (exemption for overdue actions)	
Centre Head: Please send to Gail Eagle IMAS.teaching-students@utas.edu.au to obtain Centre Head signature	
Does the project align with centre and IMAS strategic research priority areas?	
Are there sufficient funds for project?	
If no to any of the above, provide justification and return to supervisor for revision. Centre Head justification for rejection:	
Overall strategic fit: ☐ High priority / strongly meets strategy ☐ Medium priority / meets strategy ☐ Low priority	
Head of Centre Signature:	

Appendix 1

Supervisor Check list:

- Please state the expertise each supervisor brings to the project, the fraction of supervision from each supervisor for this candidate (should add up to 100%).

  "Supervisor load splits (totalling 100% for each candidate) are negotiated amongst the supervisory team and agreed by the Head of Academic Unit and should reflect percentage contribution. The Primary Supervisor must have the largest load percentage or equal share with

other members of the supervisory team. Except where approved by the Dean of Graduate Research, no supervisor may have less than 20% load share.

- A project must have a minimum of two supervisors and a maximum of four supervisors. All supervisors must be registered and additional research advisors may be added where specific expertise in one area of the project is required.
- The primary supervisor must be a contracted or permanent academic employee of the University
- The supervisory team requires at least one experienced supervisor with a minimum of 2 years supervisory experience that has supervised an HDR student to completion.
- QMS projects must have at least one UTAS and one CSIRO supervisor.
- AAPP projects: the primary supervisor must be in AAPP. This includes those funded directly by AAPP (cash-funded) or contracted/continuing IMAS staff listed to contribute into the AAPP.
- All external supervisors must be UTAS Adjunct and registered supervisors.
- Research advisors do not attract load and they do not need to be an academic staff member, registered adjunct, or registered supervisor.
- Supervisory teams require adequate academic expertise and supervisory experience to support a timely completion. Supervisory teams should aim for diversity in academic level, the development of ECRs or newly hired academics and appropriateness of supervisory loads. Gender diversity is encouraged on supervisory teams.
- Considering career stage and employment status, teams should be designed to maximise the probability that the student is well supervised throughout the candidature.

A project will not be approved if, the primary supervisor has any overdue iGrad actions, overdue confirmations or thesis submissions.

For exemption for exceptional circumstances please contact the Lead GRC Quinn Fitzgibbon quinn.fitzgibbon@utas.edu.au.

Further details about the UTAS Higher Degree by Research Supervisory Policy can be found at; http://www.utas.edu.au/__data/assets/pdf_file/0007/667213/Higher-Degree-by-Research-Supervision-Policy-December-2017.pdf